

XJTLU

IOM301

Computer-Based Trading In Financial Markets

Module Leader: Dr. Zihua LIU

Semester 2

A detailed assessment brief outline specific requirements for each assessment task. Please ensure you read it carefully **before** you undertake the assessment task.

Assessment Task 1: **Individual Work (50% of Final Mark)**

Task:	The assessment requires students to demonstrate practical skills in financial data analysis using Python through two data analytics assignments. Additionally, students will select a company or industry to produce an investment analysis report, including market and company insights, theoretical strategy design, and critical evaluation of its market-beating potential. Specific requirements see Appendix A.
Type:	Hands on Practice and Academic Report
Weight	50%
Length:	2 Financial Assignment and a Written Report (2500 words $\pm$ 500 words excluding References and Appendix).
Due Date:	Week 12, May 20th before 10am. Fail to submit on LMO, may result to fail this class. Please note Draft version on the LMO will not be accepted.
Assessment Criteria	1) Quality and accuracy of financial data extraction, processing, and analysis using Python 2) Innovation in the design of trading strategies, including novel data sources, unique analytical approaches, or insightful market observations. 3) Depth of industry/company analysis, clarity, coherence, and critical insights. 4) Application of academic theories in strategy design, evaluation of market-beating potential, and discussion of arbitrage limits. 5) Additional insightful analysis demonstrating robustness of the strategy.
Assignment Format Requirements:	1) Only submit your work in Word file. The appendix should provide a detailed description of the work performed AIGC tools 2) Name your assignment file with your studentID and name.
Reference style	1) Use of proper APA / Harvard Referencing technique, both citation and Reference. 2) Minimum of five (5) credible sources, e.g., peer review journals, industry reports, textbooks, newspapers, magazines, etc. 3) No references from Wikipedia, UK Essays, and students' papers 4) All assignments must be typed, proof-read, and professional in appearance.
Further information:	Refer to Assessment Brief on LMO



Appendix A: Task Description

This assessment aims to evaluate students' practical proficiency in financial data analytics and the creativity in designing quantitative trading strategies.

Part 1: Python Financial Data Processing Assignments

Students are required to submit two lab tutorial assignments that involve hands-on financial data analytics using Python. These assignments will test students' ability to extract, process, and analyze financial data effectively, demonstrating technical competence and applied skills in financial data manipulation.

Part 2: Industry/Company Investment Analysis and Trading Strategy Development

Students must select a specific industry or company and prepare a comprehensive investment analysis report. The report should include:

An overview of the chosen company or industry (e.g., aspects such as product/services, competitive strategies, corporate history, governance mechanisms, and marketing efforts).

A critical analysis of the company's financial data series, identifying potential undiscovered trading opportunities through an in-depth review of market dynamics, financial metrics, and relevant data.

An academic and theoretical background to the proposed trading strategy, critically evaluating its potential to outperform the market and discussing the sources of limits to arbitrage.

Optionally, students can enhance the work by conducting backtesting of proposed strategy, which will be considered a bonus score.



Appendix B: Performance Descriptors:

	Knowledge and Understanding	Intellectual Skills	Transferable Skills
100%	The best answer that could reasonably be expected from a student at that level of study under the prevailing conditions (i.e. exam or coursework)		
90-99% 'Outstanding'	Total coverage of the task set. Exceptional demonstration of knowledge and understanding appropriately grounded in theory and relevant literature.	Extremely creative and imaginative approach. Comprehensive and accurate analysis. Well- argued conclusions. Perceptive self- assessment.	Extremely clear exposition. Excellently structured and logical answer. Excellent presentation, only the most insignificant errors.
80-89% 'Excellent'	As 'Outstanding' but with some minor weaknesses or gaps in knowledge and understanding.	As Outstanding' but slightly less imaginative and with some minor gaps in analysis and/or conclusions.	As Outstanding' but with some minor weaknesses in structure, logic and/or presentation.
70-79% 'Very Good'	Full coverage of the task set. Generally very good demonstration of knowledge and understanding but with some modest gaps. Good grounding in theory.	Some creative and imaginative features. Very good and generally accurate analysis. Sound conclusions. Some self-assessment.	Generally clear exposition. Satisfactory structure. Very good presentation, largely free of grammatical and other errors.
60-69% 'Comprehensive'	As 'Very Good' but with more and/or more significant gaps in knowledge & understanding and some significant gaps in grounding.	As 'Very Good' but analysis and conclusions contain some minor weaknesses.	As 'Very Good' but with some weaknesses in exposition and/or structure and a few more grammatical and other errors.
50-59% 'Competent' Pass	Patchy coverage of the task set. Patchy knowledge and understanding with limited grounding in literature.	Rather limited creative and imaginative features. Patchy analysis containing significant flaws. Rather limited	Competent exposition and structure. Competent presentation but some significant grammatical and other

